

# **Institute of Actuaries of India**

## **Subject CT1 – Financial Mathematics**

### **April 2016 Examination**

## **INDICATIVE SOLUTION**

#### **Introduction**

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

**Solution 1:**

i)

a) Static and Dynamic hedging –

- A static hedge is the one that does not need to be re-balanced when the price or other characteristics (such as volatility) of the securities under the hedge changes.
- This contrasts with a dynamic hedge which involves regular re-balancing to maintain a hedged position at all times.

b) Arbitrage and Hedging

- Arbitrage explores market inefficiencies and involves simultaneous buying and selling of an asset in order to profit from small differences in the price.
- Hedging involves the concurrent use of more than one investment in opposite directions to limit the risk of serious investment loss.

[3]

ii) The forex trader could use the following strategy to make a profit:

- Buy a lot of EUR for \$1,079.80.
- Sell a lot of EUR for INR 74,321
- INR 74,321 could then be sold for \$1,100;

A profit of \$20 per lot can be made.

[3]

[6 Marks]

**Solution 2:**

i) An investor of a swap faces the following risks:

- Market risk – risk that the market conditions turn unfavorable and the net outgo under the agreement happens more than expected for the investor.
- Credit risk – risk of default by counterparty on its payments. This is more likely to happen if swap is of negative value to the counterparty.

[2]

ii) Yield on the corporate bonds is higher than that of the government bonds and the difference represents yield margin. Reason for increase in this margin could be:

- Security of the return on the bonds deteriorate - e.g. fall in the financial performance of the company
- Marketability of the bonds deteriorate e.g. due to over-supply of corporate bonds in the market

[2]

iii) Characteristics of property as an investment compared to the bonds:

- Property investment usually has large unit sizes leading to less flexibility than investment in bonds.

- Each property is unique & more difficult to value than the bonds
- Actual value at sale & rental income from the property is less certain than the bonds
- Buying & selling expenses on property are usually higher than bonds
- There may be a period of no income (rents) on property, while bonds mostly have a pre-defined pattern.
- Owner of the property do not face credit risk\counter party default risk like bonds.
- Marketability of property is poorer than the bonds.
- Property running yields are often lower than fixed interest bonds because of future probable capital gain.
- Expenses associated with property are higher than bonds.

[2]

[6 Marks]

**Solution 3:**

Value at 1<sup>st</sup> November 1985 is

$$\text{Annuity 1} = 200 \times v^{\left(\frac{1}{4}\right)} \times \ddot{a}_{22} \quad \text{at } 8\%$$

$$= 200 \times 0.980944 \times 11.01676 = 2161.363$$

$$\text{Annuity 2} = 320 \times (1 + i)^{\left(\frac{1}{12}\right)} \times a_{16.25}^{(4)} \quad \text{at } 8\%$$

$$= 320 \times 1.006434 \times 9.184257 = 2957.872$$

$$\begin{aligned} \text{Annuity 3} &= 180 \times a_{18.75}^{(12)} \\ &= 180 \times 9.892583 = 1780.665 \end{aligned}$$

Total value of all the three annuities = 6,899.90

Let the revised annuity be A

$$\text{Value of the revised annuity is} = A \times (1 + i)^{1/4} \times a_{21.5}^{(2)}$$

$$A = \frac{6899.90}{1.019427 \times 10.30886} = 656.56$$

[6 Marks]

**Solution 4:**

- i)  $d$  is the interest payable at the beginning of the year on a loan of 1 unit repayable at the end of the year. The corresponding interest payable at the end of year on 1 unit of loan is  $i$ .

Thus,  $d$  must equal the present value at the beginning of the year of the payment  $i$  payable at the end of the year ie  $d = iv$ .

[2]

ii) Given  $i = 0.08$

a.  $d^{(12)}$

$$\begin{aligned} v &= 1 - d = [1 - d^{(12)}/12]^{12} \\ d^{(12)} &= 12 \times [1 - v^{(1/12)}] \\ d^{(12)} &= 12 \times [1 - 1.08^{(-\frac{1}{12})}] \\ &= 0.076714776 \end{aligned}$$

b.  $i^{(365)}$

$$\begin{aligned} (1+i) &= [1 + i^{(365)}/365]^{365} \\ i^{(365)} &= 365 \times [1.08^{(\frac{1}{365})} - 1] \\ &= 0.076969155 \end{aligned}$$

c.  $\delta$

$$\begin{aligned} \delta &= \log(1 + i) \\ &= \log(1.08) \\ &= 0.076961041 \end{aligned}$$

d.  $i^{(1/2)}$

$$\begin{aligned} i^{(1/2)} &= 0.5 \times [1.08^{(2)} - 1] \\ &= 0.0832 \end{aligned}$$

[4]

(iii) From the first principles,

$$(I\ddot{a})_n = 1 + 2v + 3v^2 + 4v^3 + 5v^4 + \dots + nv^{n-1} \quad @ \mid$$

Multiplying by  $v$  on both sides,

$$v(I\ddot{a})_n = v + 2v^2 + 3v^3 + 4v^4 + \dots + nv^n$$

Subtracting,

$$(1-v)(I\ddot{a})_n = 1 + v + v^2 + v^3 + v^4 + \dots + v^{n-1} - nv^n$$

$$(1-v)(I\ddot{a})_n = (1-v^n)/(1-v) - nv^n \quad (\text{Sum of a GP formula})$$

$$d(I\ddot{a})_n = \ddot{a}_n - nv^n$$

$$(I\ddot{a})_n = \frac{\ddot{a}_n - nv^n}{d} \quad \text{Hence Proved.}$$

[2]

[8 Marks]

**Solution 5:**

i)  $v(t) = \exp \left[ - \int_0^t \delta(s) ds \right]$

Hence

$$\int_0^t \delta(s) ds = \begin{cases} 0.08t & \text{for } 0 \leq t \leq 5 \\ 0.1 + 0.06t & \text{for } 5 \leq t \leq 10 \\ 0.3 + 0.04t & \text{for } t \geq 10 \end{cases}$$

$$v(t) = \begin{cases} \exp(-0.08t) & \text{for } 0 \leq t \leq 5 \\ \exp(-0.1 - 0.06t) & \text{for } 5 \leq t \leq 10 \\ \exp(-0.3 - 0.04t) & \text{for } t \geq 10 \end{cases}$$

[2]

ii)

a) Let the lump sum value be X

The equation of value at present time is  
 $600 [v(0) + v(1) + \dots + v(14)] = X v(15)$

$$\text{Hence, } X = \frac{600 [v(0) + v(1) + \dots + v(14)]}{v(15)}$$

$$\frac{600 [1 + e^{-0.08} + \dots + e^{-0.86}]}{e^{-0.9}}$$

$$X = \text{INR } 14,119$$

b) Let A be the annuity payment. The equation of value at the present time is

$$600 [v(0) + v(1) + \dots + v(14)] = A [v(15) + v(16) + \dots + v(22)]$$

$$A = \frac{600 [1 + e^{-0.08} + \dots + e^{-0.86}]}{(e^{-0.9} + e^{-0.94} + \dots + e^{-1.18})}$$

$$A = \text{INR } 2,022$$

[6]

[8 Marks]

**Solution 6:**i) First determine the stock price at 1<sup>st</sup> June 2015,  $S_0$ . For INR 100 nominal value, evaluated at 8% p.a.:

$$S_0 = 4.5 \ddot{a}_6^{(2)} v^{4/12} + 5 \ddot{a}_6 v^{1/12} + 75 v^{5 \cdot 10/12}$$

At 8.00% p.a.;

$$\ddot{a}_6 = 4.9927$$

$$\ddot{a}_6^{(2)} = 4.898$$

$$S_0 = 4.5 \times 4.898 \times 1.08^{(-1/3)} + 5 \times 4.9927 \times 1.08^{(-1/12)} + 75 \times 1.08^{(-70/12)}$$

$$S_0 = 94.16$$

The forward price is given by  $K_0 = (S_0 - I)(1+i)^4$ , where  $I$  is income from bond and  $i = 7\%$ .

$$I = 4.5 \ddot{a}_4^{(2)} v^{4/12} + 5 \ddot{a}_4 v^{1/12}$$

$$\ddot{a}_4 = 3.624$$

$$\ddot{a}_4^{(2)} = 3.564$$

$$I = 4.5 \times 3.564 \times 1.07^{(-1/3)} + 5 \times 3.624 \times 1.07^{(-1/12)} = 33.698$$

Hence the forward price is:

$$K_0 = (S_0 - I)(1.07)^4$$

$$K_0 = (94.16 - 33.698) \times (1.07)^4 = 79.25$$

Since this is for INR 100, the forward price of contract is:

$$100 \times 79.25 = 7,925$$

[5]

ii) Value of contract on 1<sup>st</sup> August 2017 is  $(K_t - K_0) v^{10/12}$ , where  $K_t$  is forward price of contract as on 1<sup>st</sup> August 2017 with term of 1 year and 10 months;

$$K_t = (110 - I)(1+i)^{10/12}$$

$$I = 4.5 \ddot{a}_2^{(2)} v^{2/12} + 5 v^{11/12}$$

$$\ddot{a}_2^{(2)} = 1.902$$

$$I = 4.5 \times 1.902 \times 1.07^{(-1/6)} + 5 \times 1.07^{(-11/12)} = 13.16$$

$$K_t = (110 - 13.16) \times 1.07^{(22/12)} = 109.63$$

Hence, value of contract is

$$(109.63 - 79.25) \times 1.07^{(-22/12)} = 26.84$$

Since this is for INR 100, the value of contract is:

$$100 \times 26.84 = 2,684$$

[3]

[8 Marks]

### **Solution 7:**

- i) Real rate of interest is the rate of interest which will have been earned on a transaction so as to produce the total amount of cash in hand at the end of the period of accumulation reduced for the effects of inflation.

Alternate definition:-This is the interest rate achieved on a transaction allowing for the effect of inflation on the cash flows. The monetary value of the cash flow is adjusted for the inflation during the period.

[1]

ii) In periods of positive inflation, the real rate of interest will be lower than the money rate of interest. [1]

iii) If “r” is the real rate of return, “e” the inflation and “i” the money rate of return then  $(1+i) = (1+e)(1+r)$  [1]

iv) Let PV of payments made in the first 3 years be X

Note that there is no payment in the first year. There is a payment of 3,000 p.a. in the second year, 4,000 p.a. in the third year payable quarterly.

Effective rate of interest p.a. for the first 3 years =  $(1+0.08/12)^{12} - 1$   
= 0.082999507

$$X = v \cdot 3000 a_1^{(4)} + v^2 \cdot 4000 a_1^{(4)} @ 8.2999507\%$$

$$a_1^{(4)} = 0.951623562 @ 8.2999507\% \text{ hence}$$

$$\begin{aligned} X &= 0.951623562 \times [3000 \cdot v + 4000 \cdot v^2] \\ &= 0.951623562 \times 6180.469868 = 5,881.480748 \end{aligned}$$

Let PV of payments made in the second 3 years be Y

Effective rate of interest p.a. for the second 3 years =  $(1+0.12/2)^2 - 1 = 0.1236$

$$Y = v^3 @ 8.2999507\% \times [5000 a_1^{(4)} + v \cdot 6000 a_1^{(4)} + v^2 \cdot 7000 a_1^{(4)} @ 12.36\%]$$

$$a_1^{(4)} = 0.9302464865 @ 12.36\%$$

$$\begin{aligned} Y &= 0.7872546299 \cdot 0.9302464865 \cdot (5000 + 6000v + 7000v^2) @ 12.36\% \\ &= 11,632.96663 \\ Y &= 11,632.967 \end{aligned}$$

Corpus = 100,000

$$X + Y = 5,881.480748 + 11,632.967 = 17,514.447748$$

Amount available at the end of 6 years

$$\begin{aligned} &= (100,000 - 17,514.448) \cdot 1.08299^3 \cdot 1.1236^3 \\ &= \text{INR } 148,623.134 \end{aligned}$$

[6]

[9 Marks]

**Solution 8:**

- i) The time weighted rate of return for Property Fund

$$\frac{16.4}{12.4} - 1 = 0.3226 \text{ or } 32.26\%$$

The time weighted rate of return for Equity Fund

$$\frac{15.5}{12.1} - 1 = 0.2810 \text{ or } 28.10\% \quad [1]$$

ii)

- a) Assuming that the investor bought 100 units per quarter in property fund the equation of value is given by

$$1240(1+i) + 1310(1+i)^{3/4} + 1480(1+i)^{1/2} + 1580(1+i)^{1/4} = 4 \times 1640$$

$$\text{Where } i \text{ is yield per annum} = 0.2932 \text{ or } 29.32\% \quad [2]$$

- b) Assuming that the investor purchases INR 100 worth of units each quarter

$$400 \times \ddot{s}_1^{(4)} = 100 \times 16.4 \left( \frac{1}{12.4} + \frac{1}{13.1} + \frac{1}{14.8} + \frac{1}{15.8} \right)$$

$$\ddot{s}_1^{(4)} = 1.1801 \text{ this gives } i = 0.2979 \text{ or } 29.8\% \quad [2]$$

- iii) a) Assuming that the investor bought 100 units per quarter in Equity fund the equation of value is given by

$$1210(1+i) + 920(1+i)^{3/4} + 1030(1+i)^{1/2} + 1310(1+i)^{1/4} = 4 \times 1550$$

$$\text{Where } i \text{ is yield per annum} = 0.6731 \text{ or } 67.31\%$$

- b) Assume that the investor purchase INR 100 worth of units each quarter

$$400 \times \ddot{s}_1^{(4)} = 100 \times 15.5 \left( \frac{1}{12.1} + \frac{1}{9.2} + \frac{1}{10.3} + \frac{1}{13.1} \right)$$

$$\ddot{s}_1^{(4)} = 1.4135 \text{ this gives } i = 0.7088 \text{ or } 70.88\% \quad [4]$$

**[9 Marks]**

**Solution 9:**

- i) Let X be the initial annual repayment

$$160,000 = X a_{10} \text{ at } 8\%$$

$$X = \frac{160000}{a_{10}} = \frac{160000}{6.7101} = 23,844.65 \quad [2]$$

- ii) The loan outstanding immediately after the fourth payment is made is:

$$23,844.65 a_6 \text{ at } 8\% = 23,844.65 \times 4.6229 = 110,231.442$$

Let Y be the revised annual installment. The equation of value is

$$Y a_6 = 110,231.442 \text{ at } 10\%$$

$$Y = \frac{110231.442}{a_6} = \frac{110231.442}{4.3553} = 25,309.72 \quad [4]$$

- iii) The loan outstanding immediately after the seventh payment is made is

$$25,309.72 a_3 \text{ at } 10\%$$

$$= 25309.72 \times 2.4869 = 62,942.74$$

Let Z be the revised annual installment

$$Z a_3 = 62,942.74 \text{ at } 9\%$$

$$Z = \frac{62942.742}{a_3} = \frac{62942.742}{2.5313} = 24,865.77$$

The yield can be found as:

$$160,000 = 23844.65 a_4 + v^4 \times 25309.72 a_3 + v^7 \times 24865.77 a_3$$

OR

$$160,000 + 1465.07 \times a_4 - 443.95 \times a_7 - 24865.77 a_{10} = 0$$

By interpolation between 8% and 9%, the yield or effective rate of interest per annum is 8.4%

$$8\% = -4310.73$$

$$9\% = 2930.935$$

[4]  
[10 Marks]

**Solution 10:**

- i) The Above figure indicates an increasing yield curve with a slump around year 10.
- Expectation Theory

- Expectation theory states that the yield on short and long term bonds is determined by expectation of future movements in interest rates.
- An expectation of a fall in interest rates will make short term investments less attractive and longer term more attractive.
- An Excepted dip of a 10 year tenure bond in the figure above indicates that yields of long term bonds may be higher than short term bonds, implying an expectation of fall in the interest rates in the future.
- Longer term bonds will continue to give investors higher rates.
- Liquidity preference theory
  - The above curve shape is also supported by Liquidity preference theory which suggests that investors demand higher premium to compensate greater risk of loss on longer dated bonds.
  - This is because short duration bonds have lower interest rate sensitivity compared to longer dated bonds which will result in higher yield on longer duration bonds.
- Market Segmentation theory
  - Market Segmentation theory suggests term structure emerges from different forces of demand and supply of the interest rate based assets.
  - The slump at the 10 year bond tenure can be explained by Market segmentation theory. E.g. less supply of bonds from the government for a 10 year term can lead to higher prices & lower yields.
  - Securities with a term of 10 year could be more attractive for investors & they may be ready to purchase at higher prices and in return get lower yields.

[6]

ii)  $Duration = \sum t C_t v^t \setminus \sum C_t v^t$

**Duration of bond with 6% p.a. coupon rate:**

$$= \frac{6 \times (Ia)_n + 100 n v^n}{6 \times a_n + 100 v^n}$$

For n = 4 years, following are the individual items:

$$(Ia)_n = 7.9622$$

$$a_n = 3.3121$$

$$v^t = 0.73503$$

$$Duration = (6 \times 7.9622 + 100 \times 4 \times 0.73503) / (6 \times 3.3121 + 100 \times 0.73503) = 3.66 \text{ years} \quad (1.5)$$

**Duration of bond with 6.5 % p.a. coupon rate:**

$$= \frac{6.5 \times (Ia)_n + 100 n v^n}{6.5 \times a_n + 100 v^n}$$

$$6.5 \times a_n + 100 v^n$$

$$\text{Duration} = (6.5 \times 7.9622 + 100 \times 4 \times .73503) / (6.5 \times 3.3121 + 100 \times 0.73503) = 3.64 \text{ years (1.5)}$$

[3]

iii)

- The duration of a bond is the average time of cash flows weighted by present value.
- The coupon payments at 6.5% p.a will be a higher proportion of total proceeds than for at 6% p.a coupon bond.
- Thus a greater proportion of the total proceeds of 6.5% coupon bond will be received before the end of term leading to shorter duration.

[1]

**[10 Marks]****Solution 11:**

i) The cash flows of the investor:

$$\text{Coupon at 1}^{\text{st}} \text{ April 2016} = 1,000,000 \times 0.05/2 \times 23,200 / 25,200 = 23,016$$

$$\text{Coupon at 1}^{\text{st}} \text{ October 2016} = 1,000,000 \times 0.05/2 \times 25,300 / 25,200 = 25,099$$

$$\text{Coupon at 1}^{\text{st}} \text{ April 2017} = 1,000,000 \times 0.05/2 \times 26,900 / 25,200 = 26,687$$

$$\text{Redemption at 1}^{\text{st}} \text{ April 2017} = 1,000,000 \times 26,900 / 25,200 = 1,067,460$$

[2]

ii) To evaluate the purchase price on 1<sup>st</sup> October 2015, we need to determine the present value of future cash flows from the bond in money value (real terms).

Future values expressed in money value (real terms) on 1st October 2015:

$$\text{Coupon at 1}^{\text{st}} \text{ April 2016} = 23,016 * \text{RPI} (@ 1^{\text{st}} \text{ October 2015}) / \text{RPI} (@ 1^{\text{st}} \text{ April 2016})$$

$$\text{Coupon at 1}^{\text{st}} \text{ April 2016} = 23,016 \times 142 / 147 = 22,233$$

$$\text{Coupon at 1}^{\text{st}} \text{ October 2016} = 25,099 * \text{RPI} (@ 1^{\text{st}} \text{ October 2015}) / \text{RPI} (@ 1^{\text{st}} \text{ October 2016})$$

$$\text{Coupon at 1}^{\text{st}} \text{ October 2016} = 25,099 \times 142 / 154 = 23,143$$

$$\text{Coupon at 1}^{\text{st}} \text{ April 2017} = 26,687 * \text{RPI} (@ 1^{\text{st}} \text{ October 2015}) / \text{RPI} (@ 1^{\text{st}} \text{ April 2017})$$

$$\text{Coupon at 1}^{\text{st}} \text{ April 2017} = 26,687 \times 142 / 159 = 23,833$$

$$\text{Redemption at 1}^{\text{st}} \text{ April 2017} = 1,067,460 * \text{RPI} (@ 1^{\text{st}} \text{ October 2015}) / \text{RPI} (@ 1^{\text{st}} \text{ April 2017})$$

$$\text{Redemption at 1}^{\text{st}} \text{ April 2017} = 1,067,460 \times 142 / 159 = 953,329$$

The Purchase price is,

$$= 22,233 / 1.04^{.5} + 23,143 / 1.04 + 23,833 / 1.04^{1.5} + 953,329 / 1.04^{1.5}$$

$$= 965,388$$

[4]

iii) Applying the probabilities to determine the revised purchase price,

$$= 0.95 \times 22,233 / 1.04^{.5} + 0.95 \times 23,143 / 1.04 + 0.95 \times 23,833 / 1.04^{1.5} + 0.90 \times 953,329 / 1.04^{1.5}$$

$$= 886,899$$

[2]

iv) The Change in expectation of gold or inflation index will change the expected real yield from the bond. This will directly affect the price of the underlying bond and hence the price of the future.

- a) Basis an expectation of increase in the gold prices, expected return from the bond will increase. For the same inflation (retail) index level, will increase the real yield and hence the price or value of the underlying bond and future.
- b) Basis an expectation of lower inflation, for the same gold index level, will increase the real yield and hence the price or value of the underlying bond and future.

[2]

**[10 Marks]**

### **Solution 12:**

i) As  $(1 + i_t)$  is lognormally distributed, equating the formulae for mean and variance:

$$e^{\mu + 1/2\sigma^2} = E(1+i) = 1 + j = 1.1$$

$$e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) = \text{var}(1+i) = s^2 = 0.2^2$$

Solving the equations:

$$1.1^2 (e^{\sigma^2} - 1) = 0.2^2$$

$$\sigma^2 = 0.032523$$

$$\mu = \ln(1.1) - .5 \times 0.032523 = 0.07905$$

[3]

ii) Let  $S_n$  be the accumulated value after  $n$  years for an investment 1 at time 0.

$$S_n = (1+i_1) (1+i_2) \dots (1+i_n)$$

$$\log S_n = \log[(1+i_1) (1+i_2) \dots (1+i_n)]$$

$$\log S_n = \log(1+i_1) + \log(1+i_2) + \dots + \log(1+i_n)$$

Since  $\log(1 + i_t) \sim N(0.07905, 0.032523)$ , using the additive property of independent normal distributions:

$$\log S_n \sim N(n \times 0.07905, n \times 0.032523)$$

$$S_n \sim \log N(n \times 0.07905, n \times 0.032523)$$

[3]

- iii) Following the equation will help us determine probability of increase of investment INR 25,000 to be in the range of INR 32,500 and INR 35,000 in 4 years' time;

$$P(32,500 < 25,000 \times S_4 < 35,000)$$

$$P(25,000 \times S_4 < 35,000) - P(25,000 \times S_4 < 32,500)$$

Rearranging and taking log;

$$P(\ln S_4 < \ln(35,000/25,000)) - P(\ln S_4 < \ln(32,500/25,000))$$

$$\log S_4 \sim N(4 \times 0.07905, 4 \times 0.032523)$$

$$\log S_4 \sim N(0.31620, 0.13009)$$

Converting to z values:

$$P(Z < \ln((35,000 / 25,000) - 0.31620) / \sqrt{0.13009}) - P(Z < \ln((32,500 / 25,000) - 0.31620) / \sqrt{0.13009})$$

$$\Phi(0.056206) - \Phi(-0.14926)$$

$$\Phi(0.056206) = 0.522411$$

$$\Phi(-0.14926) = 1 - \Phi(0.14926) = 0.440674$$

$$= 0.522411 - 0.440674$$

Hence the probability of investment in range of INR 32,500 to INR 35,000 will be 8.17%

[4]

[10 Marks]

\*\*\*\*\*